

Spectrum-Matched Quantum-Assisted PINNs: A Constructive Design Methodology for Problem-Specific Variational Circuits, Instrumented by Explainable-AI Diagnostics

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August 7, 2026

Abstract

A classical physics-informed neural network (PINN) fails on exactly one thing, reliably and provably: it learns low frequencies fast and high frequencies almost not at all (spectral bias), because its neural tangent kernel (NTK) eigenspectrum decays with Fourier-mode index. A data-re-uploading variational quantum circuit is not a black-box speedup — it is a Fourier feature map whose accessible frequency set is exactly and prescribably determined by the eigenvalues of its data-encoding Hamiltonians. This paper joins those two facts into a constructive design rule, Spectrum-Matched Circuit Design (SMCD, Section 3): given a PDE, we compute the target spectral support analytically from the operator symbol and forcing/boundary data, then construct the encoding generators, re-uploading depth, and qubit count so the circuit’s realized frequency set covers it — no architecture search. We use six explainable-AI instruments, run identically on matched classical and hybrid models, as the measurement instrument that proves the mechanism rather than merely reporting an accuracy delta.

1 Introduction

A classical physics-informed neural network (PINN) learns low-frequency solution components fast and high-frequency components almost not at all — spectral bias — and its neural tangent kernel (NTK) eigenspectrum decays with Fourier-mode index, so residual and boundary loss terms converge at characteristically different rates. Separately, a data-re-uploading variational quantum circuit is known to realize a truncated Fourier series whose accessible frequency set is exactly determined by its data-encoding Hamiltonians’ eigenvalues [11]. These two facts have not previously been joined into a constructive design rule: prior hybrid-QPINN work [1, 3, 4, 6] reports empirical accuracy deltas on hand-picked circuit sizes, without a principled way to choose the circuit *given* the PDE.

This paper makes that connection. Given a PDE, we compute its target spectral support analytically from the operator’s Fourier symbol and forcing/boundary data, then construct the quantum circuit’s encoding generators, re-uploading depth, and qubit count so its realized frequency set covers that support — no architecture search, no hyperparameter sweep over circuit size. We call this Spectrum-Matched Circuit Design (SMCD, Section 3). We then use six explainable-AI instruments, run identically for matched classical and hybrid models at the same training checkpoints, as the measurement instrument that attributes any accuracy delta to the quantum layer specifically — not as after-the-fact decoration on a result already claimed by other means.

Contributions.

- C1 — SMCD, a constructive PDE $\rightarrow$ circuit map.** Given a linear operator’s symbol and forcing/boundary spectra, output encoding-generator eigenvalues, re-uploading depth, qubit count, and entangler pattern, deterministically. Falsifiable: a spectrum-matched circuit must beat a size-matched random-frequency circuit on the same PDE, or the design step is not earning its keep.
- C2 — Quantum-NTK block decomposition.** $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q$ for disjoint-parameter hybrid models (Section 3.5), with the quantum block’s eigenvalue decay flattened over the encoded band relative to the classical block’s power-law decay.
- C3 — Spectral-bias relief as mechanism, not just outcome.** Per-frequency error trajectories must show the hybrid model’s high- k modes converging at rates comparable to its low- k modes, with the improvement band coinciding with the encoded frequency set — not merely an aggregate accuracy improvement.
- C4 — A negative-result map with a decision rule.** An explicit characterization of PDE classes where hybridization is useless or harmful, with the mathematical reason (smoothing operators remove high-frequency content, so a high-frequency-rich circuit adds parameters and barren-plateau risk for no benefit).
- C5 — An XAI protocol for hybrid physics-informed models.** A reusable, model-agnostic instrument set with published metric definitions, run identically across classical and quantum families rather than as a post hoc, model-specific explanation.

The honest headline this paper reports, whatever the specific numbers turn out to be (Section 5–6): the quantum layer should help on high-wavenumber and multiscale problems where its encoded band matches the PDE’s spectral support, do nothing on smooth diffusive problems whose target spectrum SMCD predicts to be low-frequency-only *before any training*, and hurt when the circuit’s spectrum is mismatched to the problem. A design rule plus a stated “when not to go quantum” recommendation, adjudicated against pre-registered falsifiers (Section 5), is the contribution — not a cherry-picked win.

Assignment mapping (T5.9 self-review).

Requirement	Where it is satisfied
Solve a set of PDEs with classical PINNs and QAPINNs	Section 3 (4 PDE instances $\times$ 6 problem labels, §3.6), Section 3 (7 model families), one training harness for both
XAI techniques/metrics showing <i>how</i> learning happens in each	Section 4 (6-instrument suite), run on both families at matched checkpoints, results in Section 5
Demonstrate how the VQC affects the classical PINN learning process	Section 5 (NTK block decay, PR-7; per-frequency convergence, PR-8; matched- capacity ablation, PR-1–3), Section 6 (negative controls, PR-4, PR-11)
Methodology to construct a problem-specific circuit + architecture, with mathematical basis	Section 3 (SMCD, Propositions 1–4, Algorithm 1)
Technical report, 5–15 pp	This document (15 pp, <code>paper/main.pdf</code>)
Source code repository	<code>src/qapinn/</code> , <code>README.md</code>
Reproducibility instructions	<code>docs/REPRODUCE.md</code> , <code>README.md</code> , <code>python tasks.py repro-quick</code>
Presentation slides	<code>slides/main.pdf</code> , 18 slides, same figure assets as this document
Summary of key findings + recommendations	<code>FINDINGS.md</code> (13 numbered findings, C1–C5 table, decision table)

Self-review against this table (T5.9): every figure cited above appears in Section 5–6 with the specific prediction it decides; no verdict stated anywhere in this paper exceeds the corresponding entry in `docs/predictions.md`’s mechanical adjudication (`scripts/adjudicate_predictions.py`) — Section 8’s C1–C5 summary is a synthesis of those verdicts, not an independent claim.

2 Background

Full derivations for every item below are in `docs/derivations/`; this section states results and the one-line reason each matters to the design rule in Section 3.

2.1 Classical side

PINNs as strong-form collocation. A PINN [10] minimizes the PDE residual directly at sampled collocation points via automatic differentiation, with no mesh and no weak form — unlike Galerkin/FEM, the optimization problem is non-convex in the network parameters even when the underlying PDE is linear.

Exact derivatives via reverse-mode AD. Reverse-mode AD gives derivatives that are exact (to floating-point precision), not finite-difference approximations; the dominant cost for a PINN is second derivatives (e.g. u_{xx} in ≥ 2 dimensions), computed via double backward passes.

The NTK and lazy training [8] (Section 3.5’s classical baseline). Writing $\Theta(x, x') = \nabla_{\theta} N[u_{\theta}](x) \cdot \nabla_{\theta} N[u_{\theta}](x')^{\top}$ for the residual network’s tangent kernel, gradient-flow training linearizes in the lazy regime to $\dot{e} = -\Theta_0 e$, so each eigenmode of Θ_0 decays as $\exp(-\lambda_i t)$: loss-term imbalance between residual and boundary blocks is an eigenvalue-imbalance phenomenon, not merely a loss-weight-tuning problem (`docs/derivations/03_ntk_pinn.md`).

Spectral bias. For a tanh or ReLU MLP, the NTK eigenvalues decay roughly as a power law in Fourier-mode index, so high-frequency solution components are learned last and slowly. Fourier-feature and random-feature encodings [12] are the standard classical fix and the non-negotiable fair baseline this project compares against (model family **C-FF**, Section 5) — if the quantum layer beats only a plain MLP and not a Fourier-feature MLP, it has bought nothing that random Fourier features do not already supply.

Hard vs. soft boundary enforcement. The ansatz $u = B(x) + D(x) \cdot N(x)$ (distance function D vanishing on the boundary, particular solution B matching the boundary data) removes the boundary-loss NTK block entirely by construction, at the cost of a distance-function derivation per domain; the soft-BC alternative keeps a separately-weighted boundary loss term whose own NTK block this project instruments directly (Section 4).

2.2 Quantum side

Data-re-uploading circuits as Fourier series. For encoding gates $\exp(-ixH^{(l)})$ with generator eigenvalues $\{\lambda_a^{(l)}\}$, an L -layer re-uploading circuit’s output expectation is a truncated Fourier series whose frequency set is the set of signed sums of per-layer eigenvalue differences — the central theorem this project’s entire methodology is engineering on top of (Proposition 1, Section 3.2; 11).

Parameter-shift derivatives. $\partial_\theta f = \frac{1}{2}[f(\theta + \pi/2) - f(\theta - \pi/2)]$ for a Pauli-generated gate; because the circuit’s input x enters as a rotation angle, $\partial_x f$ is *also* exactly computable via a parameter shift — a hardware-exact PINN residual gradient, not a finite-difference approximation, and the reason this project’s whole approach is physically realizable on real quantum hardware, not just simulable.

Barren plateaus. $\text{Var}[\partial_\theta C]$ decays exponentially in qubit count n for a global-observable cost function on a deep, unstructured random circuit [9]. SMCD’s Algorithm 1 (Section 3.6) mitigates this structurally: shallow depth (by design, not by luck — the depth rule targets the *minimum* L that covers the target band), a local observable, and small-angle initialization.

Effective dimension. The Abbas et al. effective-dimension statistic [7], computed from the empirical Fisher information eigenspectrum, is this project’s quantum-native capacity measure (Section 4’s Fisher/effective-dimension instrument), reported alongside — not instead of — parameter count, since “fewer parameters” is a claim prior QPINN work makes loosely.

3 Methodology: Spectrum-Matched Circuit Design (SMCD)

3.1 Setup

The hybrid ansatz used throughout (the *serial* form; Section 5 ablates parallel and random-scaling variants) is

$$x \xrightarrow{E_\varphi} z \in \mathbb{R}^m \xrightarrow{Q_\theta} \phi(z) \in \mathbb{R} \xrightarrow{w} u(x),$$

where E_φ is an affine classical encoder, Q_θ a data-re-uploading variational quantum circuit, and w a linear head. The split is deliberate and is itself a claim: the classical part learns the coordinate *warping*, the quantum part supplies the Fourier *basis*, and the head learns the *coefficients* — exactly the structure of a spectral method with a learned coordinate map.

3.2 Proposition 1 — Circuit spectrum

Proposition 1 (Circuit spectrum). *An L -layer data-re-uploading circuit, with encoding gate $\exp(-ixH^{(l)})$ at layer l for Hermitian generator $H^{(l)} = \sum_a \lambda_a^{(l)} |a\rangle\langle a|$, realizes an observable expec-*

tation of the form

$$f(x) = \sum_{\omega \in \Omega} c_{\omega}(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L \left(\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)} \right) \right\},$$

i.e. a trigonometric polynomial whose frequency support is the set of all signed differences of encoding-generator eigenvalues, summed over layers.

Proof sketch. Expand $f(x) = \langle 0|U^{\dagger}(x, \theta) M U(x, \theta)|0\rangle$ by inserting the eigenbasis resolution of each diagonal encoding gate independently on the bra and ket sides. Every (bra-path, ket-path) pair contributes an x -dependent phase $\exp(ix \sum_l (\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)}))$ multiplied by an x -independent amplitude collecting the trainable gates and M along that path; collecting paths by total frequency gives the boxed form. The full derivation, worked first for one qubit/one layer and then generalized to n qubits and d -dimensional inputs, is in `docs/derivations/07_circuit_fourier_spectrum.md`. ■

Two subtleties that change the constructive guarantee. Two facts, easy to get wrong and verified numerically against both an independent statevector simulator and a PennyLane oracle (agreement $< 10^{-10}$) before being trusted:

1. **Generator-vs-frequency factor of two.** The generator's eigenvalues are $\pm\omega/2$ for an $RZ(\omega x)$ gate, but the *realized* frequency in $f(x)$ is the full ω , not $\omega/2$: the eigenvalue-difference structure above supplies a compensating factor of two. Reading the generator's eigenvalues off the gate and reporting that as the frequency halves every number in the design.
2. **State-preparation degeneracy.** If the qubit is prepared exactly at the equator ($RY(\pi/2)$ before the first encoding gate, an equal superposition), every frequency whose balanced-ternary leading digit is zero — exactly one third of Ω — has identically zero amplitude for *every* trainable-parameter setting, independent of ansatz expressivity. This is not a numerical inconvenience: it makes Proposition 3 below false for any target spectrum touching a dead frequency under the literal $\pi/2$ prescription. We use $\phi_{\text{prep}} = \pi/3$ throughout, verified to have no dead frequencies.

3.2.1 Corollary: ternary scaling and the corrected depth rule

Set the per-layer scalings to $\omega_l = \Delta \cdot 3^{l-1}$ (single-wire case). Then $\Omega = \Delta \cdot \{-\frac{3^L-1}{2}, \dots, \frac{3^L-1}{2}\}$ — exactly 3^L distinct frequencies with no degeneracy, a bijection with the balanced-ternary representation of the integers in that range (proof: substitute $m_l = d_l - 1$ to convert the signed sum into an ordinary base-3 expansion shifted by $\frac{3^L-1}{2}$). This grows *exponentially* in depth, versus $2L + 1$ distinct frequencies for linear scaling $\omega_l = \omega_0$ — the design knob almost every prior re-uploading paper leaves fixed.

To guarantee the reachable band covers a target maximum frequency K at ternary spacing Δ :

$$\max |\Omega| \geq K \iff \Delta \frac{3^L-1}{2} \geq K \iff L \geq \log_3 \left(\frac{2K}{\Delta} + 1 \right) \implies \boxed{L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil}.$$

This is the corrected form of the depth rule (correcting an earlier draft that stated $L = \lceil \log_3(K/\Delta) \rceil$, missing the factor of two inside the logarithm and the $+1$ term). The difference is not cosmetic: for P1's target spectrum ($K = 15\pi$, $\Delta = \pi$), the uncorrected formula gives $L = 3$, whose reachable band tops out at $13\pi < 15\pi$ — it silently misses the target mode. The corrected formula gives $L = 4$, whose band reaches $40\pi \geq 15\pi$.

3.3 Proposition 2 — PDE target spectrum

Proposition 2 (PDE target spectrum). *For a linear constant-coefficient operator $\mathcal{L}$ with Fourier symbol $\sigma(k)$, the solution of $\mathcal{L}u = f$ on a periodic/rectangular domain satisfies $\hat{u}(k) = \hat{f}(k)/\sigma(k)$. The essential spectral support $S_\varepsilon = \{k : |\hat{u}(k)| > \varepsilon \|\hat{u}\|\}$ is therefore computable before training, from $\hat{f}$ and σ alone.*

For Helmholtz, $\sigma(k) = k^2 - |\mathbf{k}|^2$, so S_ε concentrates near $|\mathbf{k}| \approx k$ — the near-resonant band a classical PINN’s low-frequency-first learning dynamics reach last, if at all. For the heat equation, σ grows quadratically with $|\mathbf{k}|$ and the solution decays exponentially in time at every nonzero mode, so S_ε collapses toward $k = 0$ as t increases: this is the project’s negative control (Section 6) — there is no high-frequency content for a quantum layer to supply, and SMCD predicts zero benefit *a priori*, not just observes it after the fact.

3.4 Proposition 3 — Matching condition and resource bound

Proposition 3 (Matching & resource bound). *If $S_\varepsilon \subseteq \Omega$, the hybrid model represents the solution to ε accuracy with a linear head — the remaining optimization problem is convex in w . Covering S_ε with maximum frequency K and resolution Δ requires $L \geq \log_3(K/\Delta)$ layers (ternary scaling, corrected form above) and $n \geq \lceil d \rceil$ qubits for d input dimensions, plus ancilla wires for any required cross-frequency terms.*

This turns architecture choice into arithmetic: given (K, Δ, d) read off the PDE’s own symbol and data, (L, n) follow by formula, not by search.

3.5 Proposition 4 — NTK consequence

Proposition 4 (Hybrid NTK decomposition). *For a hybrid model whose parameters $\theta = (\theta_{\text{cl}}, \theta_q)$ partition into disjoint classical and quantum groups,*

$$\Theta_{\text{hyb}}(x, x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x'),$$

*exactly, with **no cross term** — correcting an earlier draft of this proposition that stated $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q + 2\Theta_\times$, which is mathematically incorrect for a shared-output, disjoint-parameter model of this form.*

Proof. $\Theta(x, x') = \sum_{k=1}^P \partial_{\theta_k} f(x) \partial_{\theta_k} f(x')$ is a single inner product of gradient *vectors*, summed over the *full* index set of parameters $k = 1, \dots, P$. When that index set is partitioned into two disjoint blocks θ_{cl} and θ_q (a hybrid model’s classical and quantum parameters never overlap), the sum splits additively over the partition:

$$\Theta_{\text{hyb}}(x, x') = \sum_{k \in \text{cl}} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') + \sum_{k \in q} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x').$$

There is no third term: a cross term would only arise from a fundamentally different quantity, such as a covariance between two *separately* defined kernels or the square of a sum of outputs — not from the NTK of a single model with disjoint parameter groups, which this project’s ansatz is. This was verified independently, both analytically (the index-summation argument above) and numerically (random matched/mismatched-size synthetic Jacobians), before being trusted; `src/qapinn/xai/ntk.py` implements the corrected, no-cross-term identity. ■

Because Ω (the quantum block’s realized frequency set) is discrete and prescribed by construction, Θ_q has eigen-directions *at* the encoded frequencies with eigenvalues bounded below independently of $|n|$ for $n \in \Omega$ — the exponential/power-law eigenvalue decay in $|k|$ that drives classical spectral bias (Section 2) is replaced by a flat response over Ω . This is the formal statement of “the quantum layer removes spectral bias inside its band,” and it is what Section 5’s NTK spectrum figures measure directly: decay exponent inside Ω versus outside it, for matched classical and hybrid models.

3.6 Algorithm 1 — SMCD

Algorithm 1 Spectrum-Matched Circuit Design (SMCD)

Require: PDE operator $\mathcal{L}$, forcing f , boundary data g , domain Ω_x , tolerance ε , qubit budget $n_{\max}$

Ensure: encoding generators & scalings $\{\omega\}$, depth L , qubits n , entangler, observable, ansatz form

- 1: **Symbol:** $\sigma(k) \leftarrow$ Fourier symbol of $\mathcal{L}$ (linearize about a base state if nonlinear; Burgers uses the viscous-scale estimate $k_{\max} \sim 1/\sqrt{\nu}$)
 - 2: **Target:** $\hat{S} \leftarrow \{k : |\hat{f}(k)/\sigma(k)| > \varepsilon\} \cup$ boundary-lift spectral support (time-dependent: union over t , record growth rate)
 - 3: **Band:** $K \leftarrow \max |k|$ in $\hat{S}$; $\Delta \leftarrow$ min spacing in $\hat{S}$; $d \leftarrow \dim(x)$
 - 4: **Depth:** $L \leftarrow \lceil \log_3(2K/\Delta + 1) \rceil$ with ternary scalings $\omega_j = \Delta \cdot 3^{j-1}$ (corrected form, Section 3.2; fall back to linear scaling if $\hat{S}$ is a dense low band)
 - 5: **Width:** $n \leftarrow d + n_{\text{cross}}$, where n_{cross} covers required mixed-frequency terms (e.g. 2-D Helmholtz needs $k_x \pm k_y \Rightarrow$ entangle the two coordinate wires)
 - 6: **if** $n > n_{\max}$ or $L > L_{\max}$ (barren-plateau depth budget) **then**
 - 7: split the band into an octave ensemble (Section 3.7)
 - 8: **end if**
 - 9: **Entangler:** ring-CZ if cross terms are needed, else none (unnecessary entanglement costs gradient variance for no benefit on a separable target)
 - 10: **Observable:** local $\langle Z_0 \rangle$ (never a global Pauli string — barren plateaus)
 - 11: **Ansatz:** hard-constrained BCs, $u = B(x) + D(x) \cdot [w \cdot Q_\theta(E_\varphi(x))]$
 - 12: **Init:** small-angle $\theta \sim \mathcal{N}(0, \sigma_0^2)$, $\sigma_0 = 0.1$; identity-block init for the encoder
 - 13: **Report:** emit a design card ($\hat{S}, \Omega$, coverage = $|\hat{S} \cap \Omega|/|\hat{S}|$, L, n , param count, predicted NTK band), committed alongside every experiment run
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The design card (final step) is what makes the methodology auditable: every run’s predicted coverage can be correlated against its measured error, and that correlation — Section 5’s headline coverage-vs-error figure — is the direct empirical validation of SMCD, independent of any single problem instance’s outcome.

3.7 Octave-split circuit ensembles

When a single circuit cannot cover $\hat{S}$ within the barren-plateau depth budget ($L > L_{\max}$), SMCD falls back to M shallow circuits, each matched to one octave band $[2^j \Delta, 2^{j+1} \Delta)$ of $\hat{S}$, summed at the head. This keeps every individual circuit shallow (trainable) while the ensemble covers a wide band — multi-resolution analysis with a quantum basis per resolution level.

4 XAI Instrument Protocol

Six instruments are run at identical checkpoints ($\{0, 100, 500, 1k, 5k, 20k, \text{final}\}$ steps) for every model $\times$ PDE $\times$ seed, on matched classical and hybrid families, so any measured difference is attributable to the quantum layer and not to a difference in when or how each family was probed. Every instrument’s prediction is pre-registered (`docs/predictions.md`) before the corresponding run, with a stated falsifier — the discipline this project’s XAI protocol relies on to avoid the post-hoc-storytelling failure mode the PINN-XAI literature is criticized for [2].

7.1 NTK spectroscopy (primary). Following the template of ntk [5] for classical physics-informed solvers, extended here to the hybrid case: empirical NTK on a fixed probe set; eigenvalue spectrum, decay exponent, condition number, and per-loss-block eigenvalue mass (residual vs. boundary vs. initial condition). Predicted signature: a plateau in the hybrid model’s spectrum located at the encoded band Ω (Proposition 4, Section 3.5), against the classical model’s monotone power-law decay. NTK drift $\|\Theta_t - \Theta_0\|_F / \|\Theta_0\|_F$ is tracked to check whether the hybrid model stays in the lazy-training regime throughout.

7.2 Per-frequency error trajectory (primary). The error field $u_\theta - u^*$ projected onto Fourier modes at each checkpoint, plotted as a frequency-vs-training-step heatmap with the design card’s Ω overlaid. A classical PINN gives the textbook spectral-bias staircase (low k first); the hybrid model is predicted to show simultaneous decay across Ω . If the improved band and Ω coincide, Contribution C3 is demonstrated directly in one figure, not inferred indirectly from an aggregate error number.

7.3 Collocation-point attribution. Integrated Gradients of the residual loss with respect to input coordinates, aggregated into an attribution field over the domain, with the completeness-axiom violation and the attribution-vs-true-residual correlation reported alongside the field — Integrated Gradients is known to be unstable, so the correlation number is reported, not just the picture.

7.4 Fisher information & effective dimension. The empirical Fisher spectrum of the quantum block and its Abbas et al. effective dimension [7], tracked across training and reported against the classical block’s own effective dimension — distinguishing “more expressive” from “just more parameters.”

7.5 Gradient variance / barren-plateau tracking. $\text{Var}[\partial_\theta L]$ measured against qubit count and depth, compared to the theoretical 2^{-n} prediction for a global-cost deep random circuit [9] — the honest cost side of the design ledger.

7.6 Layerwise probes and loss-landscape slices. Ridge-regression R^2 per layer for how much of u^* is already linearly decodable at that representation, plus filter-normalized 2-D loss-landscape slices, confirming the trained point sits at a minimum.

5 Results

Sample size note, stated once here rather than repeated at every number below: the Aug 7 deadline forced the core matrix’s seed count down from the pre-registered plan of 5 to 2

(08_TASK_INDEX.md’s cut-line, invoked twice under real time pressure). At $n = 2$ paired seeds, the smallest two-sided Wilcoxon p achievable is **0.5** (both seeds agreeing in sign is the single most extreme $n = 2$ outcome), not the $n = 5$ plan’s 0.0625. docs/predictions.md’s CONFIRMED gate ($p \leq 0.0625$) was left at its original, pre-registered value rather than loosened post hoc, which means every ratio-and-significance check below (PR-1, PR-2, PR-6) can now only land on REFUTED or INCONCLUSIVE, *regardless of true effect size*. This is a real, structural consequence of the seed cut, not a weak result reported as an ambiguous one — every verdict below is exactly what the pre-registered thresholds say, unmodified.

5.1 The headline figure: coverage vs. error (PR-9)

coverage_vs_error (paper/figures/coverage_vs_error.pdf): median relative- L^2 error vs. SMCD’s achieved coverage $|\hat{S} \cap \Omega|/|\hat{S}|$, one series per problem (poisson, helmholtz_k10), Spearman ρ per problem, at $n = 18$ points per problem (6 coverage targets $\times$ 3 seeds — coverage_sweep is the one sweep never cut on seed count). **REFUTED**. Poisson shows the predicted strong negative correlation, $\rho = -0.82$ ($p = 3.2 \times 10^{-5}$), well past the $\rho \leq -0.7$ bar. Helmholtz_k10 does not: $\rho = +0.28$ ($p = 0.27$), the *wrong sign* — higher SMCD coverage did not track lower error on the harder, more quantum-heavy problem instance. The adjudication rule takes the worst case across problems, so the verdict is REFUTED even though one problem instance individually confirms the claim; this is the headline figure and it ships exactly as measured, per this project’s own “ships whatever it says” rule.

5.2 NTK spectroscopy (PR-7)

ntk_spectrum_{p1,p4,poisson_pr7,helmholtz_k10_pr7}: classical-vs-hybrid eigenvalue spectra with Ω shaded, decay exponent measured inside vs. outside the band, from each run’s committed xai/ntk_step*.npz only. **Helmholtz_k10: REFUTED**, gap = -9.67 (need $\geq +0.5$) — steeper, not flatter, inside Ω . **Poisson: INSUFFICIENT_DATA**, not REFUTED — fewer than 2 positive eigenvalues fall in the outside-band range, so that exponent is undefined, not just noisy; reported as such rather than letting $\text{NaN} \geq 0.5$ silently evaluate false. Where measurable, the quantum block’s NTK does not show the predicted eigen-concentration at the encoded frequencies.

5.3 Per-frequency error trajectories (PR-8)

freq_heatmap_{poisson,heat,burgers,helmholtz_k4,helmholtz_k10,helmholtz_k20} (all six problem instances render; see §4 note on the crash-retry specerr.npz corruption found and fixed while building this figure). **Split verdict**. Poisson: **CONFIRMED** — 71.3% of the total error improvement lands inside Ω , past the 70% bar, on real positive total improvement (13,751 in the accumulated per-frequency metric). Helmholtz_k10: **REFUTED**, and not narrowly — q_serial does not outperform c_mlp at *any* frequency (total_improvement = 0), so there is no improvement for PR-8’s premise to attribute to Ω in the first place, independent of the 70% threshold.

5.4 Ablation: is it the design, or just being quantum? (PR-1, PR-2, PR-3, PR-6)

ablation_matched: q_serial vs. c_mlp, c_ff, c_rff_matched (matched spectrum, no quantum), and q_random (quantum, unmatched spectrum), paired Wilcoxon and Cliff’s delta on Poisson (P1), $n = 2$ shared seeds. **All four REFUTED**, and in the same direction: q_serial’s median rel- L^2 (1.702) is not just short of beating the classical families, it is *worse* than every one of them at this

sample size — `c_mlp` (0.877, ratio $0.52\times$ vs. the $\geq 2\times$ bar), `c_ff` (0.115, ratio $0.068\times$ vs. the $\geq 1.3\times$ bar), and `c_rff_matched` (0.400, ratio $4.25\times$ vs. the $[0.77, 1.3]$ equivalence band PR-3 needed). PR-6 (`q_serial` vs. `q_random`, same-size ablation, C1’s own falsifier) is **INCONCLUSIVE** on Poisson (Cliff’s $\delta = 0$, the two seeds disagree in direction) and **INCONCLUSIVE** on heat despite a large surface ratio ($12.2\times$, past the $1.5\times$ bar) — Cliff’s $\delta = -1$ there means both seeds agree, but $p = 0.5$ at $n = 2$ can never clear the pre-registered significance bar (see the sample-size note above), so it is reported as inconclusive rather than confirmed on a favorable-looking ratio alone. *Adversarial check (T4.9)*: were `c_mlp`/`c_ff` simply under-tuned, faking these REFUTED verdicts? A learning-rate/width sweep, re-run at the real comparison budget, says no — `c_ff`’s mild under-tuning makes its win over `q_serial` *larger* ($18\times$, not smaller), and `c_mlp`’s apparent reduced-budget improvement was worse, not better, once tested at the actual 20000-step schedule (`docs/self_review.md`, `FINDINGS.md` F14).

5.5 The matched-capacity equivalence, across all six instances (PR-3 generalized, C1/T4.2)

PR-3’s own falsifier (`q_serial` $\approx$ `c_rff_matched`, ratio within $[0.77, 1.3]$) was checked on Poisson only in §5’s ablation above; extending the same check to all six problem instances (same $n = 2$ caveat throughout) surfaces a real pattern the single-problem view misses. Ratio (`q_serial`/`c_rff_matched` median rel- L^2): Poisson $4.25\times$ (REFUTED), heat $1.42\times$ (REFUTED), burgers $3.58\times$ (REFUTED), Helmholtz_k4 $8.97\times$ (REFUTED, driven by the same catastrophic $k=4$ non-convergence as PR-5), Helmholtz_k10 $1.09\times$ (**CONFIRMED** — inside the band), Helmholtz_k20 $0.98\times$ (**CONFIRMED**, and here `q_serial` is marginally *ahead*). The pattern is consistent with PR-5’s k -scaling trend: at low wavenumber, matched-capacity classical Fourier features decisively beat `q_serial`; at the two highest-wavenumber instances, the two are statistically indistinguishable at this sample size. This does not establish a quantum *advantage* anywhere — the best this run’s data supports is “no longer a clear quantum *disadvantage*” on the hardest two instances, which is itself a real, non-obvious finding this single-problem view would have missed.

5.6 Decision map: predicted vs. measured benefit, and P4’s k -scaling (PR-5)

`decision_map`: SMCD’s a-priori predicted benefit against the measured benefit of `q_serial` over `c_mlp`, one point per problem. Separately, P4’s k -scaling prediction (PR-5, **CONFIRMED**): the `c_mlp`/`q_serial` rel- L^2 ratio increases monotonically across $k = 4 \rightarrow 10 \rightarrow 20$ ($8.79 \times 10^{-6} \rightarrow 0.918 \rightarrow 1.024$). The direction is confirmed, but the $k = 4$ point’s extremity is itself the finding, not a clean win: a ratio of 8.79×10^{-6} means `q_serial`’s error is roughly five orders of magnitude *larger* than `c_mlp`’s at $k = 4$ — catastrophic non-convergence, not a close loss. The monotonic trend is real, but it is driven by quantum failing worst at the easiest instance and only nearly catching up at $k = 20$ (ratio 1.02, i.e. still not actually ahead), not by quantum’s advantage growing from a positive baseline.

5.7 P2’s negative control (PR-4) and the barren-plateau frontier (PR-10)

See §6 for PR-4 (heat/P2, C4’s core claim). PR-10 (gradient variance vs. the 2^{-n} barren-plateau line): **INSUFFICIENT_DATA**, reported honestly rather than computed anyway — `depth_sweep`’s final grid (cut to $n_{\text{qubits}} = 6$ only, under the same deadline pressure) has zero variation in n_{qubits} , the fit’s own independent variable, so `barren_plateau_fit` refuses to report a slope rather than return a solver artifact from a rank-deficient fit. See §7.

6 Negative Results

6.1 P2 (heat), C4’s core claim: PR-4

SMCD predicts near-zero benefit for a purely diffusive, smoothing operator *before any training run*, because $\sigma(k)$ ’s quadratic growth collapses the target spectral support toward $k = 0$ as t increases (Section 3, Proposition 2): no family should beat `c_mlp` by more than 5% on P2. Measured benefit, $(\text{median}_{\text{c_mlp}} - \text{median}_{\text{family}}) / \text{median}_{\text{c_mlp}}$, at $n = 2$ seeds: `c_ff` +24.4%, `c_rff_matched` +34.9%, `q_serial` +7.7%, `q_parallel` -67.5%, `q_random` -1023%, `q_octave` -51.2%. **REFUTED** — three families cross the 5% bar. Two of the three violators (`c_ff`, `c_rff_matched`) are classical, so they bear on “is `c_mlp` a strong baseline” generically, not on SMCD’s quantum-specific claim; `q_serial`’s +7.7% is the one genuine, if modest, violation of “hybridization is harmless-at-best on a smoothing operator.” The other three quantum families do the opposite: `q_random` in particular is an order of magnitude *worse* than `c_mlp` (-1023%), consistent with the mechanistic story (a high-frequency-rich, unmatched circuit adding parameters and barren-plateau risk with nothing for the smoothing operator to reward) — just not uniformly true of every quantum family tested, which the 5% uniform bar could not distinguish and this per-family breakdown does.

6.2 Negative-result map across all six instances

`decision_map` (§5) plots SMCD’s a-priori predicted benefit against the measured `q_serial`-vs-`c_mlp` benefit, one point per problem. Combined with §5’s ablation (PR-1/PR-2/PR-3), the honest summary at this sample size is that `q_serial` underperforms every classical family checked on P1 (Poisson) and does not clear its own falsifier against `q_random` on either P1 or P2 (PR-6, both INCONCLUSIVE at $n = 2$) — the constructive C1 claim (spectrum-matched beats size-matched-random) has no confirming evidence at the sample size this deadline allowed, not because the ratios point the wrong way (PR-6/heat’s $12.2\times$ ratio in the predicted direction) but because $n = 2$ cannot statistically separate it from chance (see §5’s sample-size note).

6.3 An unpredicted reversal: NTK drift (PR-11)

Predicted: hybrid models leave the lazy/NTK-linear training regime earlier than classical ones (larger $\|\Theta_t - \Theta_0\| / \|\Theta_0\|$ at 5k steps). Measured: median hybrid drift = 1.03, median classical drift = 6,878,097 ($n = 2$ each). **REFUTED**, and not by a small margin — classical models’ NTK moved roughly six-and-a-half million times further from initialization than hybrid models’ did. This is the sharpest reversal in the whole adjudication table and worth reading as a finding in its own right, not just a failed prediction: at this checkpoint, on this sample, it is the *classical* parameters that are decisively non-lazy, while the hybrid model’s combined (classical + quantum) NTK stayed comparatively close to its initialization. This is consistent with, though does not on its own establish, the small-angle quantum initialization ($\sigma_0 = 0.1$, Section 3) constraining the hybrid model’s effective learning dynamics near its NTK-linear regime for longer than an unconstrained classical network’s.

6.4 Encoder drift stayed small (PR-12) — the one clean methodological validation

CONFIRMED: median $\|A - I\|_F = 0.132$ at the final checkpoint, under the 0.2 bar, $n = 2$ runs. The encoder does not move far enough during training to invalidate the Ω /coverage claims computed once, at initialization (D3) — the one prediction in this table that both confirms cleanly

and underwrites the validity of every other frequency-domain claim in this paper (PR-7, PR-8, PR-9) as still describing the *trained* model, not just its untrained starting point.

7 Limitations

Stated plainly, not hedged:

1. **Statistical power, worse than planned.** The core matrix was *pre-registered* at $n = 5$ seeds per (problem, family) cell, which cannot reach $p < 0.05$ on its own (`src/qapinn/stats.py`, T4.1: the $n = 5$ Wilcoxon p -floor is 0.0625). Under the Aug 7 submission deadline, the seed count was cut to $n = 2$ (`docs/plan/08_TASK_INDEX.md`'s cut-line, invoked twice), which lowers the achievable p -floor further, to 0.5 (verified directly against `scipy.stats.wilcoxon`). `docs/predictions.md`'s CONFIRMED gate was left at its original, pre-registered 0.0625 threshold rather than loosened to compensate, which means every ratio-and-significance prediction (PR-1, PR-2, PR-6) can now only resolve to REFUTED or INCONCLUSIVE, independent of true effect size — stated here explicitly rather than left implicit in a table of verdicts.
2. **Simulation only.** All quantum circuits are evaluated on this project's own statevector simulator (`qsim.py`), cross-checked against a PennyLane oracle to $< 10^{-10}$; no results here were run on physical quantum hardware.
3. **Qubit budget.** Experiments are capped at ≤ 8 qubits, dictated by classical simulation cost, not by the design methodology itself — SMCD's depth/width formulas (Section 3) apply unchanged at larger n .
4. **Noise models are surrogates.** Shot-noise and depolarizing noise are analytic surrogates (T2.11), not a full density-matrix simulation; the discrepancy between the surrogate and a density-matrix reference on a representative subset is measured and reported (T4.4), not assumed negligible.
5. **Scope cut.** A fifth PDE (Allen–Cahn / wave, stretch goal P5) was cut from the critical path under the project's time budget; nothing in Section 3 is specific to the four PDEs actually run.
6. **The affine-encoder constraint (D3).** The classical encoder E_φ is restricted to an affine map, imposed to keep SMCD's coverage metric well-defined (a nonlinear encoder would make “the circuit's realized frequency set” itself input-dependent); this is a real restriction on the hybrid model's expressivity, not a free simplification.
7. **predicted_benefit is a heuristic.** SMCD's a-priori benefit estimate (Section 3.6's design card, plotted in `decision_map`) is a provisional heuristic, not a proven bound; Section 5 reports how well it tracks the measured benefit, and a heuristic that tracks poorly is itself a reported finding, not hidden.
8. **PR-10 (barren-plateau decay rate) is unmeasurable on this run's data, and reported as such rather than computed anyway.** `depth_sweep`'s grid was cut under the same deadline pressure to $n_{\text{qubits}} = 6$ only (dropping the originally planned $n_{\text{qubits}} = 4$ point); `barren_plateau_fit` needs at least two distinct n_{qubits} values to fit a slope, and now detects this degenerate case explicitly (`src/qapinn/xai/gradvar.py`) rather than returning

the ill-conditioned solver artifact `numpy.polyfit` would otherwise silently produce against a constant independent variable. The practical frontier (largest n before grad-var collapses below the 10^{-10} floor) is unaffected and still reported; only the decay-rate metric is the casualty.

An infrastructure limitation worth stating explicitly. The core-matrix sweep this project’s results are drawn from ran on a single consumer GPU under real memory and process-stability constraints, and required several crash-recovery restarts over the course of Phase 3. The external retry mechanism resumes correctly by run-completion (`metrics.json` existence), but a known consequence is that a small fraction of runs’ cumulative per-checkpoint diagnostic files (`specerr.npz`) were appended to across a crash boundary, producing duplicate checkpoint entries in that specific artifact (10 of 84 core-matrix runs, detected mechanically by `scripts/check_specerr_integrity.py`). Rather than exclude these runs, the duplicate entries were deduplicated post hoc (`scripts/dedupe_specerr.py`, run once all sweeps had finished so the fix could not race a still-writing task): for each of the affected files, only the last — most recent, post-crash-recovery — row for each real checkpoint step is kept, discarding the stale row written before the crash. This is a mechanical cleanup of an accumulation artifact, not new data; it recovered two figures (`freq_heatmap_poisson`, `freq_heatmap_helmholtz_k4`) and one prediction (PR-8/poisson, from INSUFFICIENT_DATA to a real CONFIRMED at 71.3% overlap) that would otherwise have been reported as missing or excluded rather than fixed.

8 Conclusion

Against the contributions claimed in Section 1, adjudicated strictly against the pre-registered falsifiers of `docs/predictions.md` (12 predictions, 15 checks counting per-problem splits; §5–6):

C1 (SMCD’s constructive map) is not supported at the achieved sample size, with one qualified exception at high wavenumber. Its own falsifier, PR-6, is INCONCLUSIVE on both checked problems — $n = 2$ seeds cannot statistically separate `q_serial` from `q_random`, even where the raw ratio points the predicted way (heat, $12.2\times$). The direct comparison against classical baselines on P1 (PR-1/PR-2/PR-3) shows `q_serial` *underperforming* `c_mlp`, `c_ff`, and `c_rff_matched` alike. Generalizing PR-3 to all six instances (§5) sharpens this: REFUTED (classical decisively wins) on four of six, but genuinely CONFIRMED — matched-capacity statistical equivalence — on the two highest-wavenumber instances, `Helmholtz_k10` and `Helmholtz_k20`, consistent with PR-5’s k -scaling trend. Read together, the honest statement is not “SMCD doesn’t work” but “this run finds no evidence of a quantum *advantage* anywhere, and finds the quantum *disadvantage* narrowing to statistical parity specifically at the hardest, highest-wavenumber instances” — a real, qualified finding, not the clean constructive win C1 was written to claim.

C2 (quantum-NTK block decomposition, flattened decay inside Ω) is reversed where measurable. PR-7 REFUTED on `Helmholtz_k10`: the quantum block’s spectrum was steeper, not flatter, inside the encoded band on this data. Poisson is INSUFFICIENT_DATA, not REFUTED — fewer than 2 positive eigenvalues fall in the outside-band index range of this run’s committed NTK spectrum, so the comparison this prediction needs is undefined there, reported as such rather than guessed at.

C3 (spectral-bias relief as mechanism) holds on one instance, not the other. PR-8 CONFIRMED on Poisson (71.3% of the real, positive improvement lands inside Ω) but REFUTED on `Helmholtz_k10`, where `q_serial` does not beat `c_mlp` at any frequency at all — there is no improvement for the mechanism to explain on the harder instance.

C4 (the negative-result map) is right about the mechanism, wrong about the uniform threshold. Three of four quantum families tested on P2 (heat) underperform `c_mlp` exactly as the smoothing-operator argument predicts, `q_random` by an order of magnitude; `q_serial`'s own modest +7.7% is the one quantum counter-example, and two classical-family violations (`c_ff`, `c_rff_matched`) are unrelated to the quantum-specific claim the 5% bar was meant to test. The mechanistic story — smoothing operators give a high-frequency-rich circuit nothing to earn its keep on — is better supported by the per-family breakdown than by the single uniform threshold PR-4 was pre-registered against.

C5 (the XAI protocol) is the contribution this run actually demonstrates cleanly. Every number above is a falsifiable, mechanically-checked quantity, run identically across classical and quantum families, computed from committed artifacts by `scripts/adjudicate_predictions.py` rather than asserted — and the protocol caught real bugs in its own supporting tooling along the way (a NaN silently miscategorized as REFUTED in PR-8's early implementation, a degenerate least-squares fit that would have reported a fabricated PR-10 slope, corrupted checkpoint data recovered rather than silently used), which is itself evidence the falsification discipline this paper argues for is not decorative.

The honest headline, matching Section 1's own framing: at the sample size this deadline allowed ($n = 2$, not the pre-registered $n = 5$), this run does not demonstrate the quantum-advantage hypothesis SMCD was built to test — on the one problem with a full ablation (Poisson), the hybrid model underperforms every classical comparator, and the NTK-block and spectral-flattening mechanisms proposed to explain *why* it should help (C2) point the opposite direction from predicted. What does hold up is the negative-result mechanism on the smoothing control (C4, mostly), one genuine per-frequency win (C3, Poisson only), and the measurement protocol itself (C5) — which is precisely the outcome this project's own pre-registration discipline exists to make legible rather than obscure. A larger, unrushed replication at the originally planned $n = 5$ seeds, and a `depth_sweep` grid that retains n_{qubits} variation (needed to even attempt PR-10), are the two most direct next steps (Section 7).

9 Reproducibility appendix

Every number below is measured against this session's actual runs, not estimated.

- **Hardware:** NVIDIA RTX 4090 Laptop GPU, 32 logical CPU cores (`provenance.json`, recorded per run) — one machine for every run in this paper.
- **Software** (`docs/ENV_RESOLVED.md`, pinned via uv): Python 3.12.13, torch 2.11.0+cu128 (CUDA 12.8), pennyLane 0.45.1, numpy 2.4.4, scipy 1.18.0.
- **Determinism:** `qapinn.seeding.set_global_seed` seeds Python, NumPy, and Torch (CPU+CUDA) from one integer and enables `torch.use_deterministic_algorithms`; an explicit `torch.Generator` is threaded through every sampling call, so two runs at the same seed produce bit-identical `metrics.json` (T0.20).
- **Seeds used** (state explicitly, cut from a pre-registered $n = 5$ under the Aug 7 deadline — Section 7, FINDINGS.md F10): `core_matrix {0,1}`; `coverage_sweep/alpha_sweep/noise_study/soft_bc_ntk {0,1,2}`; `depth_sweep {0}`.
- **Concurrency:** `--workers 2 --mem-fraction 0.35` for every sweep except `depth_sweep` (`--workers 1 --mem-fraction 0.8`, forced down after system-memory exhaustion on its

larger circuits — OS/driver contention under concurrent CUDA context creation on this specific machine, not a VRAM limit).

- **Measured wall-clock:** `tasks.py repro-quick` (CPU-only, 42 combinations, implemented this session): **513.7s** (≈ 8.6 min), 42/42 OK, under the 15-min target. `tasks.py test` (384 tests): 361.8s. `tasks.py figures` (18 figures from committed data, no GPU): under a minute. A single composite `repro-all` time was not measured — that command is not implemented as one entry point this session; per-experiment costs are in `results/cost_ledger.json` instead.

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